

The L^q , $p = 1$ Endpoint via Poissonization

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Status

This packet gives a candidate partial solution to Remark 12.10 of Yaroslavl'tsev's paper *Local characteristics and tangency of vector-valued martingales* [1]. It is partial relative to the source's broad UMD-space problem, but it closes the full $p = 1$ endpoint for the concrete class $L^q(S)$, $1 < q < \infty$.

Earlier packets in this run prove the $L^q(S)$ -valued case for $1 < p, q < \infty$ and prove the $p = 1$ endpoint for $1 < q \leq 2$ by a Bernstein-function argument. The new contribution here is the missing $q > 2$ square-function comparison, proved by a Poissonization equivalence for positive $L^{q/2}$ -valued sums.

Source Problem

The source is arXiv:1907.11588, PDF page 105, Remark 12.10. The source states that the general local-martingale case is open and reduces the missing issue to an independent-increment comparison.

...the fact that M and N are quasi-left continuous. $\square$

Remark 12.10. It is not known whether Theorem 12.8 holds for general local martingales. By Theorem 3.39 and by Proposition B.1 the main issue here is in proving a similar statement for discrete martingales with independent increments. Let us state this problem here as open. Let X be a Banach space, $1 \leq p < \infty$. Let $(\xi_n)_{n \geq 1}$ and $(\eta_n)_{n \geq 1}$ be X -valued mean-zero independent random variables such that for any Borel $B \in X \setminus \{0\}$

$$\sum_n \mathbb{P}(\xi_n \in B) \leq \sum_n \mathbb{P}(\eta_n \in B).$$

Does there exist a constant C (perhaps depending on p and X) such that

$$\mathbb{E} \left\| \sum_n \xi_n \right\|^p \leq C \mathbb{E} \left\| \sum_n \eta_n \right\|^p?$$

The continuation states the aggregate-domination formulation and notes that even the symmetric case was not known to the author.

By a standard symmetrization trick [66, Lemma 6.3] one can assume that ξ_n 's and η_n 's are symmetric. But even the symmetric case is not known for the author.

Main Results

For a finite independent family $x = (x_i)$ of $L^q(S)$ -valued random variables, put

$$\mathrm{SF}_q(x) = \mathbb{E} \left\| \left(\sum_i |x_i|^2 \right)^{1/2} \right\|_{L^q(S)}.$$

Let

$$\mu_x = \sum_i \mathcal{L}(x_i)|_{L^q(S) \setminus \{0\}}$$

be the aggregate occurrence measure.

Theorem 1 (Endpoint square-function comparison). *Let $1 < q < \infty$. There is a constant C_q such that for every measure space S and every two finite independent families $x = (x_i)$, $y = (y_j)$ of $L^q(S)$ -valued random variables,*

$$\mu_x \leq \mu_y \implies \mathrm{SF}_q(x) \leq C_q \mathrm{SF}_q(y).$$

Theorem 2 (General local L^q , $p = 1$). *Let $1 < q < \infty$, let S be a measure space, and put $X = L^q(S)$. If N and M are X -valued local martingales and N is characteristically dominated by M in the sense of [1, Definition 12.9], then*

$$\mathbb{E} \sup_{t \geq 0} \|N_t\|_{L^q(S)} \leq C_q \mathbb{E} \sup_{t \geq 0} \|M_t\|_{L^q(S)}.$$

Idea of the Proof

For $1 < q \leq 2$, the previous packet proves Theorem 1 by a positive multivariate Bernstein representation. For $q > 2$, put

$$r = q/2 > 1, \quad \alpha = 1/2.$$

The square-function comparison becomes a comparison for independent positive $L^r(S)$ -valued variables:

$$\mathbb{E} \left\| \sum_i V_i \right\|_{L^r}^\alpha \lesssim_r \mathbb{E} \left\| \sum_j W_j \right\|_{L^r}^\alpha$$

under aggregate occurrence-measure domination.

The key new observation is that this expectation is equivalent to the same expectation for the compound-Poisson positive sum whose intensity is the aggregate occurrence measure. The compound-Poisson sum depends only on that aggregate measure. Therefore aggregate domination becomes immediate: if $\nu \leq \mu$, the Poisson sum with intensity μ is the sum of the Poisson sum with intensity ν and an independent positive remainder.

Thinning Lemmas

Throughout this section $r > 1$, $\alpha = 1/2$, and

$$\Phi(f) = \|f\|_{L^r(S)}^\alpha, \quad f \in L^r(S)_+.$$

The functional Φ is monotone on the positive cone and subadditive.

Lemma 1 (Scalar positive Rosenthal). *For every $r > 1$ there is a constant C_r such that for all finite independent nonnegative scalar random variables $Z_1, \dots, Z_n$,*

$$\mathbb{E} \left(\sum_i Z_i \right)^r \leq C_r \left[\left(\sum_i \mathbb{E} Z_i \right)^r + \sum_i \mathbb{E} Z_i^r \right].$$

Proof. For $1 < r \leq 2$ this follows from the von Bahr–Esseen inequality applied to $Z_i - \mathbb{E} Z_i$, plus the triangle inequality in L^r . For $r > 2$, the usual scalar Rosenthal inequality gives

$$\mathbb{E} \left| \sum_i (Z_i - \mathbb{E} Z_i) \right|^r \leq C_r \left[\left(\sum_i \mathbb{E} Z_i^2 \right)^{r/2} + \sum_i \mathbb{E} Z_i^r \right].$$

Let $A = \sum_i \mathbb{E} Z_i$ and $B = \sum_i \mathbb{E} Z_i^r$. By interpolation,

$$\mathbb{E} Z_i^2 \leq (\mathbb{E} Z_i)^{(r-2)/(r-1)} (\mathbb{E} Z_i^r)^{1/(r-1)}.$$

Hölder's inequality gives

$$\sum_i \mathbb{E} Z_i^2 \leq A^{(r-2)/(r-1)} B^{1/(r-1)}.$$

Raising to $r/2$ and using Young's inequality yields

$$\left(\sum_i \mathbb{E} Z_i^2 \right)^{r/2} \leq A^r + B.$$

Combining this with $\|\sum_i Z_i\|_r \leq A + \|\sum_i (Z_i - \mathbb{E} Z_i)\|_r$ proves the lemma. $\square$

Lemma 2 (Upper thinning). *Let $0 \leq a \leq 1$. Let $f_1, \dots, f_n \in L^r(S)_+$ be deterministic, and let $R_1, \dots, R_n$ be independent Bernoulli variables with $\mathbb{P}(R_i = 1) \leq a$. Then*

$$\mathbb{E}_R \Phi \left(\sum_i R_i f_i \right) \leq K_r a^{1/(2r)} \Phi \left(\sum_i f_i \right).$$

Proof. Let $\rho_i = \mathbb{P}(R_i = 1)$. By Lemma 1, applied pointwise in $s \in S$,

$$\mathbb{E}_R \left(\sum_i R_i f_i(s) \right)^r \leq C_r \left[\left(\sum_i \rho_i f_i(s) \right)^r + \sum_i \rho_i f_i(s)^r \right].$$

Since $\rho_i \leq a$ and $f_i \geq 0$,

$$\left(\sum_i \rho_i f_i \right)^r \leq a^r \left(\sum_i f_i \right)^r, \quad \sum_i \rho_i f_i^r \leq a \sum_i f_i^r \leq a \left(\sum_i f_i \right)^r.$$

Integrating and using $a^r \leq a$ gives

$$\mathbb{E}_R \left\| \sum_i R_i f_i \right\|_{L^r}^r \leq C'_r a \left\| \sum_i f_i \right\|_{L^r}^r.$$

Jensen's inequality for the concave map $t \mapsto t^{\alpha/r}$ yields the claim. $\square$

Lemma 3 (Lower thinning above one half). *Let $f_1, \dots, f_n \in L^r(S)_+$ be deterministic. Let $R_1, \dots, R_n$ be independent Bernoulli variables with $\mathbb{P}(R_i = 1) = c$, where $1/2 \leq c \leq 1$. Then*

$$\Phi \left(\sum_i f_i \right) \leq 2 \mathbb{E}_R \Phi \left(\sum_i R_i f_i \right).$$

Proof. By subadditivity,

$$\Phi \left(\sum_i f_i \right) \leq \Phi \left(\sum_i R_i f_i \right) + \Phi \left(\sum_i (1 - R_i) f_i \right).$$

Taking expectations, the second term has the same law as a Bernoulli $(1 - c)$ thinning. Since $1 - c \leq c$, this thinning can be coupled as a sub-thinning of a Bernoulli c thinning. Monotonicity of Φ gives

$$\mathbb{E} \Phi \left(\sum_i (1 - R_i) f_i \right) \leq \mathbb{E} \Phi \left(\sum_i R_i f_i \right),$$

and the result follows. $\square$

Poissonization

Let $V_1, \dots, V_n$ be independent $L^r(S)_+$ -valued variables. Write

$$p_i = \mathbb{P}(V_i \neq 0).$$

When $p_i > 0$, write Z_i for a random variable with the conditional law of V_i given $V_i \neq 0$; when $p_i = 0$, omit that index. Let

$$\Pi_V = \sum_i \sum_{k=1}^{N_i} Z_{i,k},$$

where N_i are independent Poisson variables with means p_i , and $(Z_{i,k})$ are independent copies of Z_i . Equivalently, Π_V is the compound-Poisson positive sum whose intensity measure is

$$\nu_V = \sum_i \mathcal{L}(V_i)|_{L_+^r \setminus \{0\}}.$$

Proposition 1 (Positive L^r Poissonization). *For every $r > 1$, there is $A_r < \infty$ such that*

$$A_r^{-1} \mathbb{E} \Phi \left(\sum_i V_i \right) \leq \mathbb{E} \Phi(\Pi_V) \leq A_r \mathbb{E} \Phi \left(\sum_i V_i \right).$$

Proof. First prove the lower bound. Let $c = 1 - e^{-1} > 1/2$. Since

$$\mathbb{P}(N_i \geq 1) = 1 - e^{-p_i} \geq c p_i \quad (0 \leq p_i \leq 1),$$

the first nonzero Poisson layer stochastically dominates a c -thinning of the original family. More explicitly, if $V_i = B_i Z_i$ with B_i Bernoulli p_i , and if R_i are independent Bernoulli c variables, then

$$\sum_i B_i R_i Z_i$$

can be coupled below $\sum_i \mathbf{1}_{\{N_i \geq 1\}} Z_{i,1}$, which is itself dominated by Π_V in the positive cone. Conditional on (B_i, Z_i) , Lemma 3 gives

$$\Phi \left(\sum_i B_i Z_i \right) \leq 2 \mathbb{E}_R \Phi \left(\sum_i B_i R_i Z_i \right).$$

Taking expectations and using monotonicity gives

$$\mathbb{E} \Phi \left(\sum_i V_i \right) \leq 2 \mathbb{E} \Phi(\Pi_V).$$

For the upper bound, decompose the Poisson sum into layers:

$$\Pi_V = \sum_{k \geq 1} L_k, \quad L_k = \sum_i \mathbf{1}_{\{N_i \geq k\}} Z_{i,k}.$$

By subadditivity,

$$\mathbb{E} \Phi(\Pi_V) \leq \sum_{k \geq 1} \mathbb{E} \Phi(L_k).$$

For $0 \leq p \leq 1$,

$$\mathbb{P}(\text{Pois}(p) \geq k) \leq \frac{p^k}{k!} \leq \frac{p}{k!}.$$

The first inequality follows from $\mathbf{1}_{\{N \geq k\}} \leq \binom{N}{k}$ and $\mathbb{E} \binom{N}{k} = p^k/k!$ for $N \sim \text{Pois}(p)$. Thus L_k has the law of a thinning of the original nonzero family with retention probabilities at most $a_k = 1/k!$. Applying Lemma 2 conditionally to the original family gives

$$\mathbb{E} \Phi(L_k) \leq K_r (1/k!)^{1/(2r)} \mathbb{E} \Phi \left(\sum_i V_i \right).$$

Since $\sum_{k \geq 1} (1/k!)^{1/(2r)} < \infty$, summing over k proves the upper bound. $\square$

Theorem 3 (Positive aggregate comparison for $q > 2$). *Let $q > 2$, $r = q/2$. Let V_i, W_j be finite independent families of positive $L^r(S)$ -valued random variables. If*

$$\sum_i \mathcal{L}(V_i)|_{L_+^r \setminus \{0\}} \leq \sum_j \mathcal{L}(W_j)|_{L_+^r \setminus \{0\}},$$

then

$$\mathbb{E} \left\| \sum_i V_i \right\|_{L^r}^{1/2} \leq C_q \mathbb{E} \left\| \sum_j W_j \right\|_{L^r}^{1/2}.$$

Proof. Let $\nu_V \leq \nu_W$ be the two aggregate occurrence measures. Construct compound-Poisson sums Π_V, Π_W . Since $\nu_W = \nu_V + (\nu_W - \nu_V)$, Π_W has the same law as $\Pi_V + \Pi'$, where Π' is an independent positive compound-Poisson sum. Monotonicity of Φ gives

$$\mathbb{E}\Phi(\Pi_V) \leq \mathbb{E}\Phi(\Pi_W).$$

Apply Proposition 1 to both families. □

Proof of the Square-Function Theorem

Proof of Theorem 1. For $1 < q \leq 2$, this is exactly the theorem proved in the companion packet

1907.11588_Lq_p1_endpoint_ql2_square_function.

Let $q > 2$ and set $r = q/2$. Given $L^q(S)$ -valued variables x_i, y_j , put

$$V_i = |x_i|^2, \quad W_j = |y_j|^2.$$

The map $x \mapsto |x|^2$ is continuous from $L^q(S)$ to $L^{q/2}(S)$, so aggregate domination $\mu_x \leq \mu_y$ pushes forward to aggregate domination of the positive L^r -valued square magnitudes. Theorem 3 gives

$$\mathbb{E} \left\| \sum_i |x_i|^2 \right\|_{L^r}^{1/2} \leq C_q \mathbb{E} \left\| \sum_j |y_j|^2 \right\|_{L^r}^{1/2},$$

which is precisely $\text{SF}_q(x) \leq C_q \text{SF}_q(y)$. □

Discrete and General-Local Consequences

Proposition 2 (Discrete $p = 1$ comparison). *Let $1 < q < \infty$. Let $(\xi_i)_{i=1}^n$ and $(\eta_j)_{j=1}^m$ be independent mean-zero $L^q(S)$ -valued random variables. If*

$$\sum_i \mathbb{P}(\xi_i \in B) \leq \sum_j \mathbb{P}(\eta_j \in B) \quad (B \in \mathcal{B}(L^q(S)), 0 \notin B),$$

then

$$\mathbb{E} \max_{1 \leq k \leq n} \left\| \sum_{i=1}^k \xi_i \right\|_{L^q(S)} \leq C_q \mathbb{E} \max_{1 \leq k \leq m} \left\| \sum_{j=1}^k \eta_j \right\|_{L^q(S)}.$$

Proof. The aggregate hypothesis is $\mu_\xi \leq \mu_\eta$, so Theorem 1 compares the two lattice square functions. Yaroslavtsev's Banach-function-space BDG theorem [2, Theorem 4.32 and Proposition 4.33] says that for every $L^q(S)$ -valued martingale difference sequence $d = (d_i)$,

$$\mathbb{E} \sup_N \left\| \sum_{i=1}^N d_i \right\|_{L^q} \simeq_q \mathbb{E} \left\| \left(\sum_i |d_i|^2 \right)^{1/2} \right\|_{L^q}.$$

Both independent mean-zero families are martingale difference sequences in any fixed order. Applying this equivalence to (ξ_i) and (η_j) proves the maximal comparison. □

Proof of Theorem 2. The conditional endpoint packet in this run proves that Theorem 1, for a fixed q , implies the full $L^q(S)$ -valued general-local characteristic-domination theorem at $p = 1$. For completeness we recall the reduction.

Yaroslavtsev's decoupled tangent construction and canonical decomposition reduce the comparison to continuous parts and purely discontinuous tangent jump martingales. Continuous parts are controlled at $p = 1$ by the source's continuous characteristic-domination estimate [1, Proposition 12.6]. For the jump part, conditional on the local characteristics and after the usual truncation of time and jump size, the decoupled tangent jumps are finite independent mean-zero families. Characteristic domination controls the total spatial jump compensator,

$$\nu^N(\mathbb{R}_+ \times B) \leq \nu^M(\mathbb{R}_+ \times B), \quad B \in \mathcal{B}(L^q(S)), \quad 0 \notin B,$$

which is exactly the aggregate occurrence-measure domination in Proposition 2.

Apply Proposition 2 conditionally and pass through the truncation and localization limits. Combining the discontinuous estimate with the continuous part and the $p = 1$ Banach-function-space BDG/canonical-decomposition equivalence [2, Section 4.3] gives the zero-initial-value case. Nonzero initial values are handled by the initial-value domination in Definition 12.9 and the triangle inequality. $\square$

Verification and Novelty Notes

The new argument is self-contained apart from scalar Rosenthal's inequality and Yaroslavtsev's $p = 1$ Banach-function-space BDG theorem. The Montgomery-Smith–Pruss comparison theorem [5] was used as a guide for looking at iid and Poisson aggregate models, but the Poissonization proposition above is proved directly for the positive L^r -lattice functional needed here.

Cheap run indexes contained prior packets for $1 < p$ in L^q , Hilbert all p , ℓ_q at $p = q$, the $1 < q \leq 2$ endpoint, a deterministic $q > 2$ subcase, the broad c_0 counterexample, and the conditional endpoint reduction. No packet proved the full $L^q(S)$, $p = 1$, $1 < q < \infty$ endpoint.

Human Review Recommendation

Review Proposition 1 first. The lower bound uses only the fact that $1 - e^{-p} \geq (1 - e^{-1})p$ and the lower thinning lemma. The upper bound is the only delicate step: decompose Poisson counts into layers and use the upper thinning lemma with the summable factors $(1/k!)^{1/q}$. Once that proposition is accepted, aggregate domination follows immediately by Poisson superposition.

References

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